

Introduction to **GWAS**

Data Pre-Processing: Initial & Exploratory Data Analysis

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Initial data analysis (IDA) mainly focuses on data cleaning, a first screening, and transformation (if necessary) to ensure **data quality** and **confirm that our data set meets the relevant distributional and model assumptions**. Data cleaning may include steps such as elimination of duplicate records, handling of missing values, identification of systematic errors, or correction of coding inconsistencies.

Exploratory data analysis (EDA) is used to examine data sets and summarize their main characteristics. EDA helps to **discover patterns in the data, spot anomalies and outliers, test a hypothesis and check our assumptions**. EDA tells us what data can reveal beyond the formal modeling or hypothesis testing and provides a better understanding of data set variables and their interactions. It can also help to determine if the statistical techniques you are considering for data analysis are appropriate.



Initial and Exploratory Data Analysis

("crap in, crap out")

Initial and Exploratory Data Analysis – IDA & EDA

Before we conduct a GWAS, there are **two types of data** to explore

- **Genotypic data**
- **Phenotypic data**

Initial and Exploratory Data Analysis – IDA & EDA

Genotypic data

IDA & EDA - **genotypic data**

Various metrics – statistics are performed either **across SNP** or **across samples** (individuals)

Key concept: detect **SNP and samples that should be removed** prior to GWAS

Use of some metrics and thresholds is species- and population-specific

Still some level of subjectivity in the thresholds

IDA & EDA - **genotypic data**

Also referred to as Quality Control (QC)

Some parameters to look at...

- Genotype calling and signal intensities (not covered here; bioinformatics topic, often automated)
- Marker allele frequencies
- Missing rate per marker and per individual
- ~~Hardy Weinberg equilibrium~~
- Heterozygosity
- Marker distribution across the genome

IDA & EDA - **genotypic data**

Missing rate per marker & per individual

SNPs might be of poor quality if their genotyping failed in many individuals

- Should be investigated separately for all study groups (if known)
- Common SNP array thresholds are 2 – 10% (based on sample size & SNP number)

Sample DNA might be of poor quality if there are many missing SNPs in an individual

- Too many missing SNPs per individual can be an indication of poor DNA quality
- (or true deletions...)
- Common SNP array thresholds are between 2 – 10% (based on sample size & SNP number)
- Includes monomorphic SNP !!

IDA & EDA - genotypic data

Marker allele frequencies

- Allele counts & genotype counts
- Minor allele frequency (MAF)
 - Some SNPs will be monomorphic
 - One of the alleles may be at very low frequency (or count)
 - Might be due to genotyping errors (genotyping has become very accurate, though!)
 - Power to detect the association is very low

Common MAF thresholds are between 1 – 3% (but think about what's reasonable in your data set!)

In samples with known group structure, MAF should be checked within groups

IDA & EDA - genotypic data

Minor allele count - a (better) alternative to MAF

- Many genotyping platforms have become extremely accurate and may not justify a filtering based on a “relatively high” expected genotyping error rate anymore.
- Minor allele count is based on the idea that you need a certain number of samples (alleles) to reliably estimate the mean effect of the allele.
- “Think about it in terms of a t -test” - you want to check if two alleles have a different effect on a trait.
- **Rule of thumb**: 30-60 counts of the minor allele

Exploratory data – genotypes

Marker	Individuals												
	1	2	3	4	5	6	7	8	9	10	11	12	13
2	2	2	2	2	2	2	2	2	2	2	2	2	2
2	2	2	2	2	2	-1	2	2	2	2	2	2	2
2	-1	2	2	2	2	2	1	0	2	1	2	2	2
2	2	2	2	2	2	2	2	2	2	2	2	2	2
0	1	0	0	0	0	-1	0	0	0	0	0	0	0
0	0	0	0	0	0	-1	0	0	0	0	0	0	0
2	1	1	0	0	0	-1	0	0	1	0	0	0	1
0	2	1	1	1	2	2	1	1	2	0	1	1	2
1	0	1	0	0	0	0	0	0	0	0	0	0	0
1	2	1	1	1	2	2	1	2	2	1	2	1	1

Exploratory data – genotypes

Monomorphic marker

Marker	Individuals												
	1	2	3	4	5	6	7	8	9	10	11	12	13
1	2	2	2	2	2	2	2	2	2	2	2	2	2
2	2	2	2	2	2	-1	2	2	2	2	2	2	2
3	2	-1	2	2	2	2	1	0	2	1	2	2	2
4	2	2	2	2	2	2	2	2	2	2	2	2	2
5	0	1	0	0	0	-1	0	0	0	0	0	0	0
6	0	0	0	0	0	-1	0	0	0	0	0	0	0
7	2	1	1	0	0	-1	0	0	1	0	0	0	1
8	0	2	1	1	2	2	1	1	2	0	1	1	2
9	1	0	1	0	0	0	0	0	0	0	0	0	0
10	1	2	1	1	2	2	1	2	2	1	2	1	1

Exploratory data – genotypes

Marker	Individuals												
	1	2	3	4	5	6	7	8	9	10	11	12	13
	2	2	2	2	2	2	2	2	2	2	2	2	2
	2	2	2	2	2	-1	2	2	2	2	2	2	2
	2	-1	2	2	2	2	1	0	2	1	2	2	2
	2	2	2	2	2	2	2	2	2	2	2	2	2
	0	1	0	0	0	-1	0	0	0	0	0	0	0
	0	0	0	0	0	-1	0	0	0	0	0	0	0
	2	1	1	0	0	-1	0	0	1	0	0	0	1
	0	2	1	1	2	2	1	1	2	0	1	1	2
	1	0	1	0	0	0	0	0	0	0	0	0	0
	1	2	1	1	2	2	1	2	2	1	2	1	1

Monomorphic marker

Low MAF

Exploratory data – genotypes

Monomorphic marker

Individuals

Marker	2	2	2	2	2	2	2	2	2	2	2	2	2
	2	2	2	2	2	-1	2	2	2	2	2	2	2
	2	-1	2	2	2	2	1	0	2	1	2	2	2
	2	2	2	2	2	2	2	2	2	2	2	2	2
	0	1	0	0	0	-1	0	0	0	0	0	0	0
	0	0	0	0	0	-1	0	0	0	0	0	0	0
	2	1	1	0	0	-1	0	0	1	0	0	0	1
	0	2	1	1	2	2	1	1	2	0	1	1	2
	1	0	1	0	0	0	0	0	0	0	0	0	0
	1	2	1	1	2	2	1	2	2	1	2	1	1

Low MAF

Many missing markers

Exploratory data – genotypes

Monomorphic marker

Individuals

Marker

2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
2	2	2	2	2	-1	2	2	2	2	2	2	2	2	2
2	-1	2	2	2	2	1	0	2	1	2	2	2	2	2
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
0	1	0	0	0	-1	0	0	0	0	0	0	0	0	0
0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0
2	1	1	0	0	-1	0	0	1	0	0	0	0	1	1
0	2	1	1	2	2	1	1	2	0	1	1	1	2	2
1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
1	2	1	1	2	2	1	2	2	1	2	1	1	1	1

Low MAF

Many missing markers

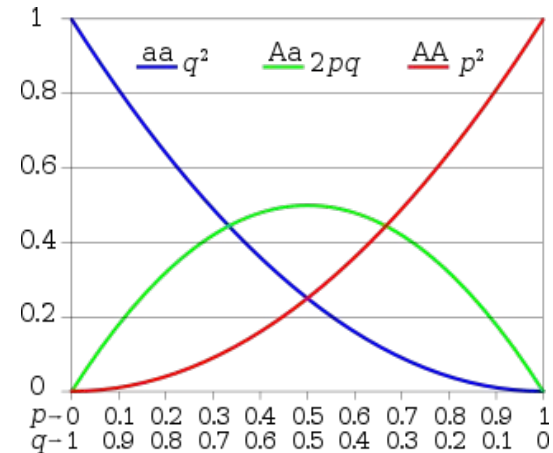
No SNP with too many missing calls removed!

IDA & EDA - genotypic data

Hardy-Weinberg equilibrium: relationship between allele and genotype frequencies

$$(p + q)^2 = p^2 + 2pq + q^2$$

- diploid genomes
- autosomal loci
- large population
- random mating
- equal frequencies in both sexes
- no selection
- no migration
- no mutations



IDA & EDA - **genotypic data**

Deviations from HW equilibrium

- systematic genotyping errors
- violation of assumptions

Test for HW deviation

- chi-squared (χ^2) test
- Fisher's (exact) test
- many, many more ...

But also selection, assortative mating, population structure and inbreeding cause deviations from HWE!

HWE is, in most cases, NOT a reasonable assumption...

IDA & EDA - genotypic data

Heterozygosity

- Proportion of heterozygotes
- Heterozygosity can be checked per locus & per individual

Very high sample heterozygosity can be an indication of DNA contamination

- But also could be that a small proportion of samples are truly very different from the rest...
- Removal of samples that depart ± 3 SD from the mean

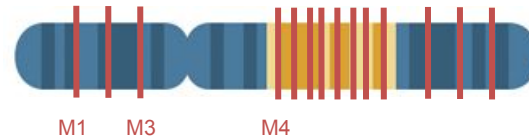
Very high heterozygosity per marker could also indicate poor DNA quality, but also be due to...

- the breeding scheme (e.g. hybrid breeding in plants, or very low heterozygosity in lines)
- Genome duplications

IDA & EDA - **genotypic data**

Linkage Disequilibrium

- Optimally, markers are relatively equally distributed across the genome
→ Regions with high recombination rate are more densely covered
- Irregular distribution of markers can result in an over- or underestimation of the true relationship between individuals



IDA & EDA - **genotypic data**

Linkage Disequilibrium - LD pruning

... in PLINK

[Linkage disequilibrium - PLINK 1.9](#)

... in R

[Tutorials for the R/Bioconductor Package SNPRelate](#)

... in Tagger

[Tagger: Selection and evaluation of tag SNPs](#)

Summary of pre-processing for genotype data

Pre-processing genotypic data – **Standard procedures**

missing rate

- per-sample (e.g. max 10% missing SNP genotypes per sample)
- per-site (e.g. max 5% missing genotype per variant/SNP)
- stricter/looser thresholds depending on data/experiment (e.g. SNP array, GBS, quality of reference sequence, ...)

MAF

Hardy Weinberg equilibrium

others

Pre-processing genotypic data – Standard procedures

missing rate

MAF

- remove monomorphic variants → non-informative
- remove variants at low frequency (“rare”) → spurious associations
- threshold may depend on sample size
- usually (re)done after imputation
- consider using minor allele count instead!

Hardy Weinberg equilibrium

others

Pre-processing genotypic data – Standard procedures

missing rate

MAF

Hardy Weinberg equilibrium

- set **low threshold** for p-value (e.g. $\exp(-10)$)
- **questionable**: some of the forces driving HW disequilibrium are characteristic for breeding (selection, migration, mutation, adaptation etc.)

others

Pre-processing genotypic data – Standard procedures

missing rate

MAF

Hardy Weinberg equilibrium

others

- heterozygosity
- marker distribution (LD pruning)
- sex chromosomes (might need to be removed / analyzed separately)
- relatedness (between samples - check for duplicates)
- Mendelian errors (unexpected inheritance patterns)

IDA & EDA - **genotypic data**

Phenotypic data

IDA & EDA - phenotypic data

Data type

- Continuous (e.g. height)
- Binary (e.g. case/control)
- Categorical (e.g. scores (ordered), eye colour (ordered))

Measure of centrality: mean, mode, median

Measures of dispersion: range, variance, standard deviation

Distribution of the data

- Distribution of values as expected? Outliers?
- representative sample of the population?
- Other explanatory covariables

IDA & EDA - phenotypic data

Covariates

Are there any variables which may have a relationship with the phenotype?

- E.g. sex, breed, age, treatments, year effect, ... (population structure)

The data needs to be corrected for these effects. Otherwise they can be confounded with allelic variants with an effect on the phenotype which we try to identify.

Covariates with significant effects on the phenotype can be identified using ANOVA (requires balanced datasets - ANOVA is outdated) or **linear mixed models**. However, a comprehensive preparation of phenotypic data including model comparison is not covered here...

IDA & EDA - **phenotypic data**

Assumptions for continuous variables (different for binary traits...)

- Normally distributed **residuals** (prerequisite of GWAS model assumptions)
- Homogeneity of variance (differences in variance might indicate a factor that has not been included in the phenotype processing)

IDA & EDA - phenotypic data

Outliers

- Apparently rare phenotypes are often a result of errors or poor models rather than true outliers
- However, the values might be real - outliers should be investigated thoroughly rather than relying on statistical tests

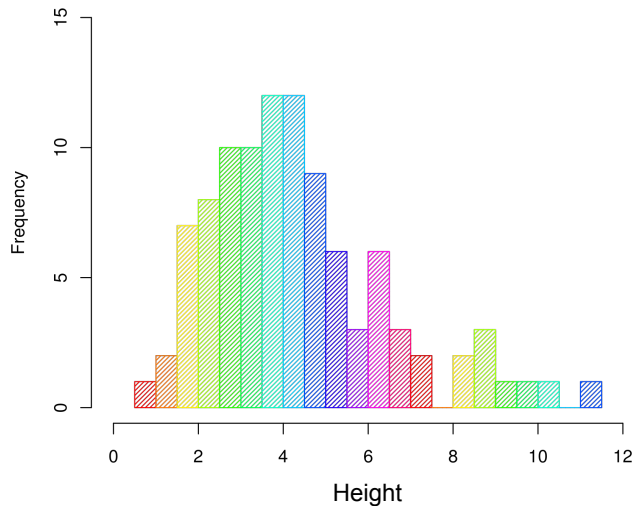
Data transformation

- **Positively skewed distributions of the residual** with the long tail into the positive direction can be corrected with a logarithmic or square root transformation.
- **Negatively skewed distributions of the residual** that have a long tail in the negative direction can be corrected with cubing or squaring
- Transformations only if really necessary...

IDA & EDA - phenotypic data

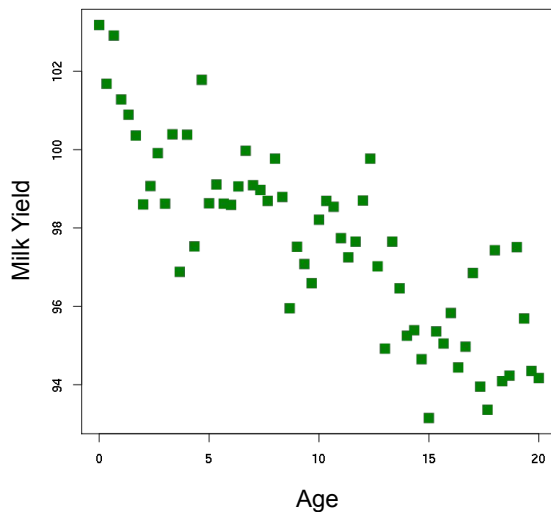
Histograms

Histogram of arrivals

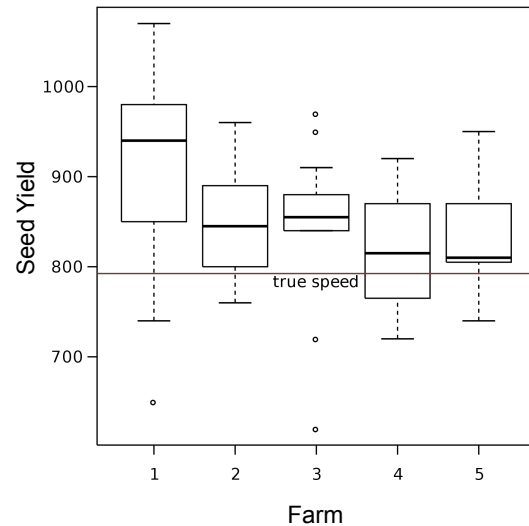


Scatter plots

Scatterplot for quality characteristic XXX



Box plots



IDA & EDA - phenotypic data

Normality of residuals

- Look at the data (Histogram, QQ-Plot)
- Don't rely on Shapiro-Wilk test

Variance homogeneity

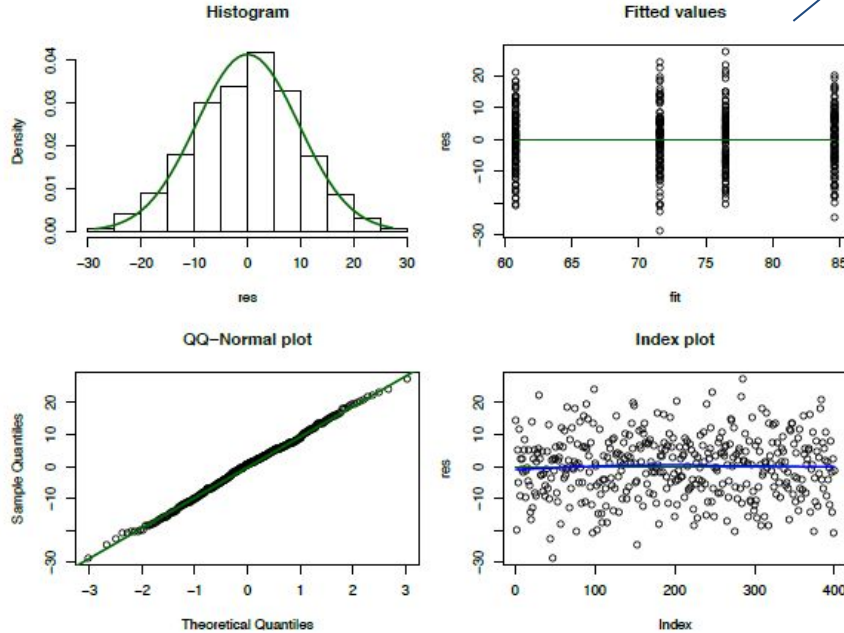
- Look at the data (Scatterplot)
- Don't rely on Leven's test

Tests are conservative and might indicate a violation of the assumptions of normality and variance homogeneity in a suitable “real-world” dataset.

IDA & EDA - phenotypic data

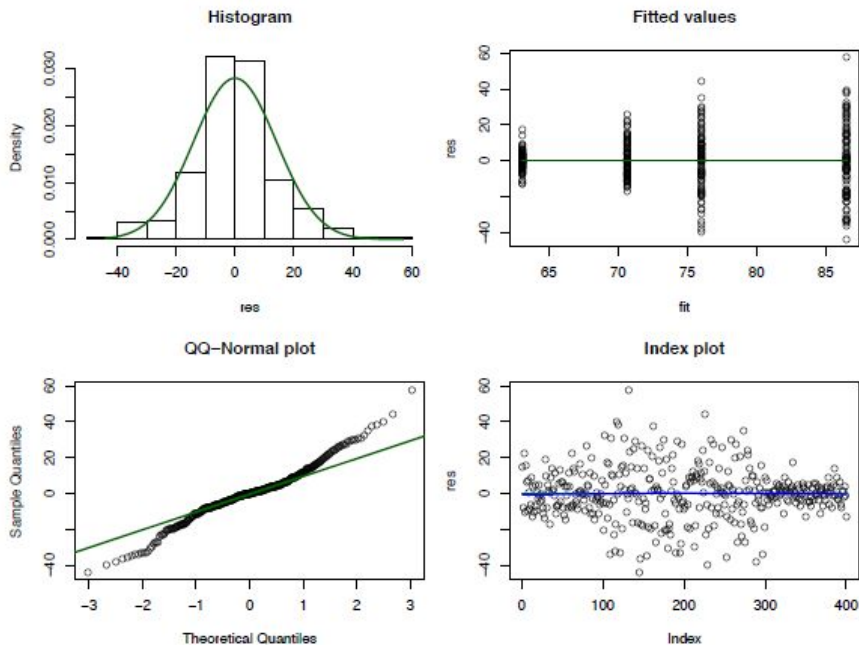
Almost perfectly distributed **residuals**

Fixed effect with 4 levels
Similar residual distribution within each factor level.



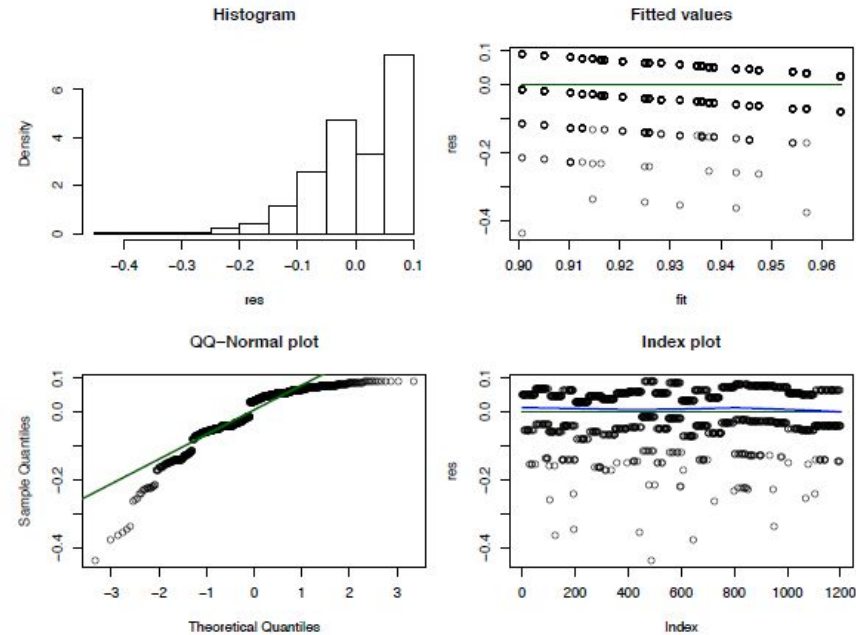
IDA & EDA - phenotypic data

Still good...



IDA & EDA - phenotypic data

There might be something wrong here....



Imputation of missing genotypes

Imputation of missing genotypes – **why?**

Imputation - the process of replacing missing data with substituted values

Preliminary step for a wide range of genetic analyses

Most models and software for population genetics, genomic selection (GS) and genome-wide association studies (GWAS) **do not handle missing data** by default and require complete datasets

1. Genotyping techniques generate a proportion of missing data (uncalled genotypes)

- SNP arrays ~5%
- RAD-Seq (e.g. GBS) ~50%

2. Optimization/efficiency of genotyping strategies (low → high density data)

scaling-up: **low** → **high density** (mixed genotyping strategies)

whole-genome sequence imputation

Imputation of missing genotypes – **methods**

1. **General methods for the imputation of any type of data**

- mean substitution (replacing missing values with the mean of the SNP across the population), median imputation
- K-Nearest Neighbour Imputation (KNNI)
- many more ...

2. **Methods specific for the imputation of missing genotypes**

Two groups (and combinations of them):

- based on LD and allele frequency (Beagle)
- based on pedigree information